Homework

1. Solve $dy/dx = 4xy^{1/2}$ with $y(0) = 1$ using the linear multi-step method

   $y_{i+2} - y_{i+1} = (1/3)\, h\, (3f_{i+1} - 2f_i)$

   First, find the exact solution for the given problem. Then, make use of the exact solution for any additional starting values in order to apply the linear multi-step method. Compare your numerical solution against the exact solution. Explain your observation.

2. Solve $dy/dx = y - x$ with $y(0) = 1$ using the linear multi-step method
   $y_{i+1} + (3/2)y_i - 3y_{i-1} + (1/2)y_{i-2} = 3hf_i$.

   First, find the exact solution for the given problem. Then, make use of the exact solution for any additional starting values in order to apply the linear multi-step method. Compare your numerical solution against the exact solution. Explain your observation.

3. Construct a linear multi-step method to solve $dy/dx = f(x,y)$ with $y(0) = y_0$ as follows. First, find a polynomial approximation in x for the function f(x,y) assuming that three points (x_i,f_i), (x_{i-1},f_{i-1}) and (x_{i-2},f_{i-2}) are known. Then, substitute this approximation for f(x,y) in the problem and integrate it between x_i and x_{i+1}. Find the order of this method.